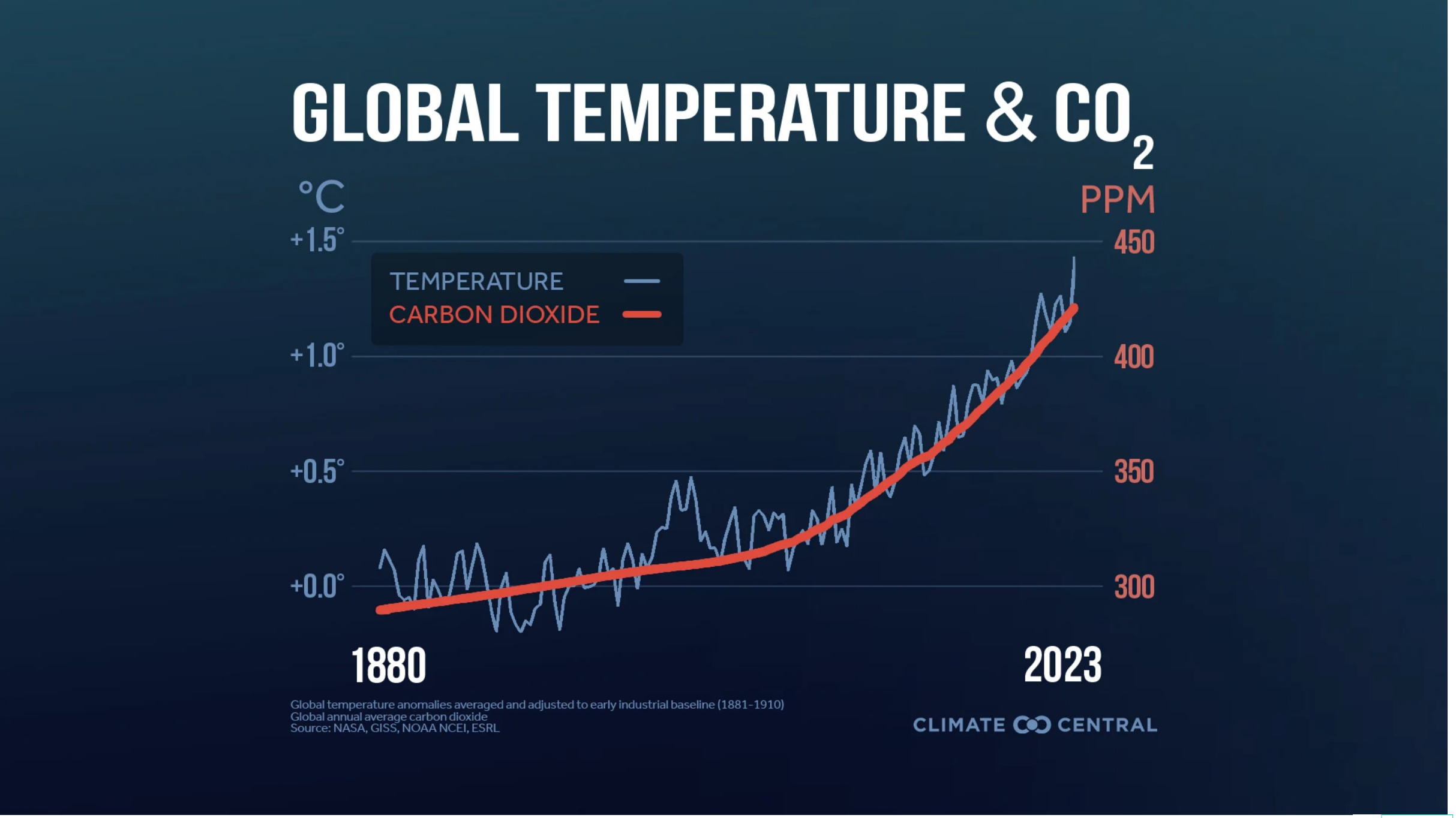
# 2 Data Summary and Presentation

# GLOBAL TEMPERATURE

- **Analysis Tools**: Data can be analyzed using both numerical summaries and graphical techniques.
- **Best Practice**: Visual methods are essential; every strong statistical analysis must begin by **plotting the data**.


GLOBAL TEMPERATURE & CO2
°C
PPM
+1.5°
+1.0°
+0.5°
+0.0°
450
400
350
300
TEMPERATURE
CARBON DIOXIDE
1880
2023
Global temperature anomalies averaged and adjusted to early industrial baseline (1881-1910)
Global annual average carbon dioxide
Source: NASA, GISS, NOAA NCEI, ESRL
CLIMATE CENTRAL

# CHAPTER OUTLINE

- **Data Overview**: Techniques for summarizing and displaying data (**2-1**).
- **Diagrams & Histograms**: Detailed methods for **stem-and-leaf diagrams** (**2-2**) and **histograms** (**2-3**).
- **Box & Time Series Plots**: Using **box plots** (**2-4**) for distribution and **time series plots** (**2-5**) for trends.
- **Multivariate Data**: Analysis and visualization of **multiple variables** simultaneously (**2-6**).

## LEARNING OBJECTIVES for this chapter:

- **Statistical Calculations**: Learn to calculate and interpret key data points like the **mean**, **variance**, **standard deviation**, **median**, and **range**.
- **Conceptual Differences**: Understand the distinctions between **sample** statistics and **population** parameters.
- **Visual Displays**: Build and analyze data visualizations such as **stem-and-leaf displays**, **histograms**, and **box plots** to identify data patterns.

- **Comparative Analysis**: Use **box plots** and other graphical tools to compare different data sets side-by-side.

- **Time & Relationships**: Utilize **time series plots** for trend analysis and **scatter plots** (along with **correlation coefficients**) to evaluate relationships between variables.

# 2-1 DATA SUMMARY AND DISPLAY

- Data features can be described **numerically**. Key ways to characterize these features include:

  - **Central Tendency**: This represents the "location" or center of the data.
  - **Arithmetic Average**: The standard method used to find the mean.
  - **Sample Mean**: Since data is typically viewed as a subset of a larger group, the arithmetic average is referred to as the **sample mean**.

- The **Sample Mean** is the arithmetic average of a set of observations. Its calculation is summarized below:

  - **Observations**: Represented as $x_1, x_2, \ldots, x_n$ for a sample of size $n$.

  - **Calculation**: The **sample mean** ($\bar{x}$) is found by summing all observations and dividing by the total number of items.

  - **Formula**: $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$ **(2-1)**

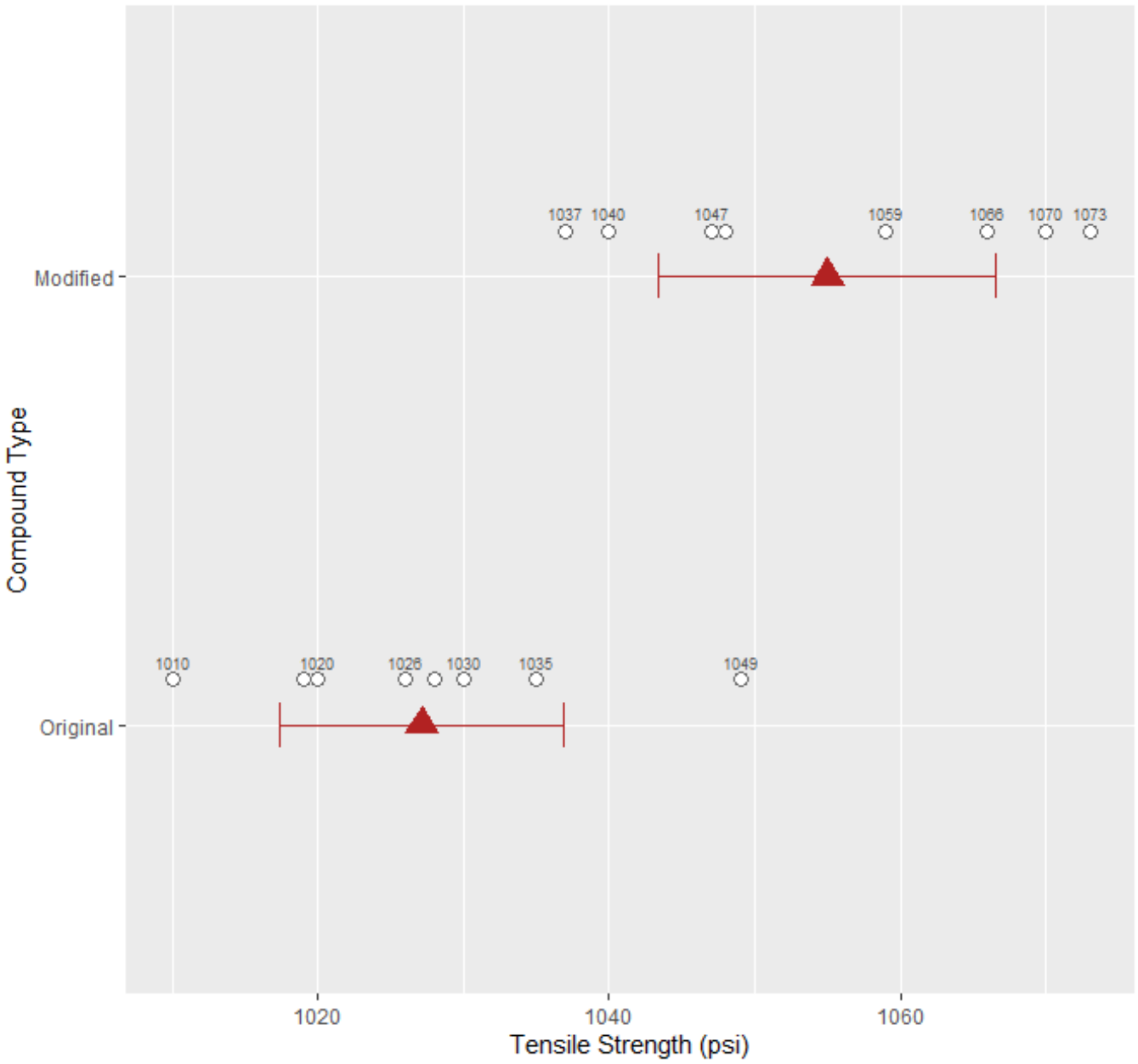
## EXAMPLE 2-1 O-Ring Strength: Sample Mean

- **Experiment**: An analysis of tensile strength for a modified rubber compound used in O-rings.
- **Visual Representation**: The data is displayed in a **dot diagram** (**Fig. 2-2**).
- **Calculation**: The sample mean for the eight observations is calculated as:

$$\bar{x} = \frac{8440}{8} = 1055.0 \text{ psi}$$

Figure 2-2: O-Ring Tensile Strength Dot Diagram

Triangles indicate Sample Means; horizontal lines show 95% CI



- We calculate averages for different groups of data:

  - **Sample Mean:** The average value calculated from a specific **sample** of observations.

  - **Population:** The larger, overall group of items or observations from which a **sample** is taken (e.g., all O-rings sold or a factory lot of wafers).

  - **Population Mean:** The average value of every single observation within the entire **population**, represented by the Greek letter $\mu$.

- **Population Mean Formula:** For a finite population of size $N$, the calculation is $\mu = \sum_{i=1}^{N} x_i / N$, identified as equation **(2-2)**.

- **Sample Mean** Utility: The **sample mean** ($\bar{x}$) serves as a practical estimate for the true population mean ($\mu$).

- **Application:** In the O-ring study, the engineer uses the **sample mean** of 1055 psi to estimate the overall average tensile strength.

- While the sample mean provides a central value, it doesn't show the full picture. To understand how spread out or scattered the data is, we use:

  - **sample variance**
  - **sample standard deviation**

- **Sample Variance**: Calculated as $s^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2/(n-1)$, identified as equation **(2-3)**, the **sample variance** measures the spread of $n$ observations.

- **Sample Standard Deviation**: The **sample standard deviation** ($s$) is the positive square root of the **sample variance**.

- **How Does the Sample Variance Measure Variability?**: This measure tracks dispersion by looking at the deviations ($x_i - \bar{x}$) of data points from the mean.

- **Fig. 2-3** Visualization: As shown in **Fig. 2-3**, larger absolute deviations indicate greater variability in the data set, such as the O-ring tensile strength.

- **Squaring Deviations**: Since deviations always sum to zero, the **sample variance** squares them to ensure all values are non-negative.

- **Interpreting Results**: A small $s^2$ indicates low variability, while a large $s^2$ signifies that the data points are more spread out.

- **EXAMPLE 2-2 O-Ring Strength: Sample Variance**: This example demonstrates the calculation of variability using tensile strength data.

- **Table 2-1** Calculations: As shown in **Table 2-1**, the sum of the squared deviations $\sum(x_i - \bar{x})^2$ equals 1348.

- **Fig. 2-3** Plot: The data points and their deviations from the mean are visualized in the dot diagram in **Fig. 2-3**.

- **Results**: Using the formula, the sample variance $s^2$ is $1348/7 = 192.57$ psi$^2$, and the sample standard deviation $s$ is $\sqrt{192.57} \approx 13.9$ psi.

- The **simple computing formula** for **sample variance** ($s^2$) and **standard deviation** ($s$) provides an efficient alternative to calculating deviations ($x_i - \bar{x}$), which can be tedious and prone to rounding errors with non-integer data. The **computational formula**, defined as $s^2 = \dfrac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1}$ **(2-4)**, is frequently referred to as the **shortcut method** because it involves squaring individual observations and their sum directly.

- **EXAMPLE 2-3 O-Ring Strength: Alternative Variance Calculation**: This exercise focuses on finding the sample variance and standard deviation for O-ring tensile strength.

- **Methodology**: It employs the "shortcut method" (Equation 2-4) rather than the standard definitional formula.

- **Sample Variance Calculation**:

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1} = \frac{1348}{7} = 192.57 \text{ psi}^2$$

- **Standard Deviation Calculation**:
  $s = \sqrt{192.57} = 13.9 \text{ psi}$

- **Conclusion**: The shortcut method yields the exact same results as previous calculation methods, confirming its reliability.

- **Population Variance**: Represented by the Greek letter $\sigma^2$ (sigma squared), this measures the spread of all values within a population.

- **Population Standard Deviation**: Represented by $\sigma$, this is the positive square root of the variance.

- **Formula for Finite Populations**: For a population of $N$ values with a mean of $\mu$, the variance is calculated as:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \quad \textbf{(2-5)}$$

- **Divisor Differences**: The **sample** variance uses $n-1$ as the divisor, while the population variance uses the total population size $N$.

- **Role of the Mean**: Since the true population mean ($\mu$) is usually unknown, the **sample** average ($\bar{x}$) is used to calculate deviations.

- **Correction Factor**: Observations in a **sample** tend to be closer to $\bar{x}$ than to $\mu$. Using $n-1$ instead of $n$ compensates for this tendency.

- **Avoiding Bias**: If $n$ were used as the divisor for a **sample**, the resulting variability would be consistently smaller than the true population variance.

- **Conceptual Basis**: The sample variance $s^2$ is considered to be based on $n-1$ **degrees of freedom**.
- **Zero Sum Constraint**: Because the $n$ deviations from the mean always sum to zero, only $n-1$ values can vary independently.
- **Automatic Determination**: Once $n-1$ deviations are known, the final value is fixed, meaning only $n-1$ **degrees of freedom** exist.

- **Visual Evidence**: This relationship between the deviations and the **degrees of freedom** was previously demonstrated in **Table 2-1**.